

Requirements Traceability Matrix

ID	Requirement	Sprint / Design Artifact	Test Case(s)
FR-1	Display company hierarchy as an interactive organizational tree.	Sprint 4 – Org Chart Visualization (D3.js integration)	TC-1.1 Expand/collapse nodes TC-1.2 Click opens employee profile
FR-2	Employee profile must display: name, phone number, desk, department, position, manager, direct reports.	Sprint 5 – Employee Detail & Department Views	TC-2.1 Profile fields populate correctly TC-2.2 Missing fields handled gracefully
FR-3	Search by employee name, position, or department.	Sprint 6 – Search and Filtering	TC-3.1 Search by name TC-3.2 Search by department/position TC-3.3 Response <2s
FR-4	Department view listing all employees within a department.	Sprint 5 – Employee Detail & Department Views	TC-4.1 Department page lists all employees TC-4.2 Links navigate to profiles
FR-5	Navigation between profiles (manager ↔ employee ↔ direct reports).	Sprint 5 – Employee Detail & Department Views	TC-5.1 Manager → Employee TC-5.2 Employee → Direct report
FR-6	CRUD operations for employees and departments.	Sprint 2 – Backend API Design	TC-6.1 Create/Update/Delete employee/department

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			TC-6.2 Org chart updates dynamically
FR-7	Authentication with enterprise credentials (SSO).	Sprint 7 – Security and Offline Support	TC-7.1 Login with valid SSO TC-7.2 Block unauthorized logins
FR-8	Role-based authorization to restrict sensitive fields.	Sprint 7 – Security and Offline Support	TC-8.1 Standard users see only permitted fields TC-8.2 Admins see all
FR-9	Offline access to cached org chart and employee data.	Sprint 7 – Security and Offline Support (IndexedDB/PouchDB)	TC-9.1 Access cached data offline TC-9.2 Sync updates on reconnect
FR-10	Error handling for invalid input, failed searches, and server downtime.	Sprint 2/6 – Backend & Search handling	TC-10.1 Invalid input error TC-10.2 Search no results TC-10.3 Downtime fallback message
FR-11	Integration with cloud-hosted database (MongoDB/DynamoDB).	Sprint 2 – Backend API Design	TC-11.1 Data stored/retrieved correctly TC-11.2 DB connection failure handled
FR-12	Reporting structure must reflect HR data.	Sprint 5 – Employee Detail & Department Views	TC-12.1 Manager/direct report matches HR records

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NFR-1	Org chart/profile views load in <2s (95%).	Sprint 4/5 – Performance Optimization	TC-N1.1 Load testing
NFR-2	Search queries return in <2s.	Sprint 6 – Search and Filtering	TC-N2.1 Benchmark search API
NFR-3	Support 10,000 employees & 500 concurrent users.	Sprint 8 – Refactoring / Load testing	TC-N3.1 Stress test dataset TC-N3.2 Simulate concurrent users
NFR-4	All communication HTTPS/TLS; data encrypted at rest.	Sprint 7 – Security and Offline Support	TC-N4.1 TLS verification TC-N4.2 DB encryption audit
NFR-5	Auth & authorization logged, auditable.	Sprint 7 – Security and Offline Support	TC-N5.1 Audit logs verify all access
NFR-6	99.9% uptime.	Sprint 9 – Testing, Polishing, Deployment	TC-N6.1 Monitor cloud uptime
NFR-7	Cross-browser, mobile responsive.	Sprint 5 – Employee Detail & Department Views	TC-N7.1 Test Chrome/Edge/Firefox/Safari TC-N7.2 Test mobile/tablet
NFR-8	WCAG 2.1 AA compliance.	Sprint 5 – Employee Detail & Department Views	TC-N8.1 Accessibility audit passed
NFR-9	≥80% unit test coverage; CI/CD pipelines.	Sprint 8 – Refactoring	TC-N9.1 Automated test coverage reports

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NFR-10	Compliance with enterprise identity providers/security standards.	Sprint 7 – Security and Offline Support	TC-N10.1 Security review sign-off
NFR-11	Hosted in AWS/Azure/GCP with secure configuration.	Sprint 9 – Testing, Polishing, Deployment	TC-N11.1 Deployment validated TC-N11.2 Security configuration audit